

Reviewer Pack — Minimal Local Index of Kähler Manifolds and Circuit Entangle...

Entry cbd4f788bc0d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Index of Kähler Manifolds and Circuit Entanglement Inequality

Entry ID: cbd4f788bc0d **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 19:16:26 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Kähler Geometry) **Field B** (complexity object): Quantum Information Theory (Circuit Entanglement)

Statement:

For every n -vertex circuit C with entanglement $E(C)$, the minimal local index of its Kähler metric $I(K)$ is such that $I(K) \leq c * E(C)$, where c is a constant.

Rationale (proposer's reasoning):

Kähler geometry provides a framework for understanding geometric structures in complex analysis, which may reveal intrinsic properties of quantum information processing tasks like circuit entanglement. The local index captures the complexity of these geometric structures, potentially providing a bridge between classical geometry and quantum information theory.

Taxonomy category: ALGEBRAIC_GEOMETRY_X_QUANTUM_INFORMATION_THEORY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 10512e26c981bb5f

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 95% of the generated n -vertex circuits C with increasing entanglement $E(C)$, the ratio $I(K)/E(C)$ converges to a value within $\pm 10\%$ of c .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Kähler manifolds" AND "circuit entanglement inequality" - "minimal local index" AND ("quantum information theory" OR "circuit complexity") - "Algebraic Geometry" AND "entanglement in quantum computation"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = 10 # Starting size for circuits
    c = 2.0 # Hypothetical constant from the conjecture

    total_ratio = 0.0
    instances_tested = 0
    n_max = 0

    while instances_tested < 30:
        entanglement = random.uniform(1, 10) * n # Simulating increasing entanglement
        kahler_index = random.uniform(1, 2) * n # Simulating Kähler metric's minimal local index

        ratio = kahler_index / entanglement
        total_ratio += ratio
        instances_tested += 1
        n_max = max(n_max, n)

    if instances_tested >= 30:
        break

    n += 5

    mean_ratio = total_ratio / instances_tested
    conjecture_holds = abs(mean_ratio - c) <= 0.1 * c
    counterexample = "" if conjecture_holds else f"Mean ratio {mean_ratio} does not converge to {c}"

    return {
        "metric_name": "ratio",
```

```

        "metric_value": mean_ratio,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, **{trial_result}}}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) /
                           len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if support_fraction >= 0.95:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE insufficient_data")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'counterexample': 'Mean ratio 0.3502946798832723 does not converge to 2.0'}}
TRIAL: {"seed": 421, **{'metric_name': 'ratio', 'metric_value': 0.4373087111212118, 'instances_tested': 1000000}}
TRIAL: {"seed": 463, **{'metric_name': 'ratio', 'metric_value': 0.4078248601982024, 'instances_tested': 1000000}}
TRIAL: {"seed": 503, **{'metric_name': 'ratio', 'metric_value': 0.332967125436039, 'instances_tested': 1000000}}
TRIAL: {"seed": 547, **{'metric_name': 'ratio', 'metric_value': 0.285789099780216, 'instances_tested': 1000000}}
TRIAL: {"seed": 593, **{'metric_name': 'ratio', 'metric_value': 0.3543735734796731, 'instances_tested': 1000000}}
TRIAL: {"seed": 631, **{'metric_name': 'ratio', 'metric_value': 0.37292121502134756, 'instances_tested': 1000000}}
TRIAL: {"seed": 677, **{'metric_name': 'ratio', 'metric_value': 0.34392967389825496, 'instances_tested': 1000000}}
TRIAL: {"seed": 727, **{'metric_name': 'ratio', 'metric_value': 0.3670912511189293, 'instances_tested': 1000000}}
TRIAL: {"seed": 773, **{'metric_name': 'ratio', 'metric_value': 0.41476600220448684, 'instances_tested': 1000000}}

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The ratio I(K)/E(C) does not converge to a value within $\pm 10\%$ of c for all tested instances, indicating that the conjecture is false. | next: Investigate the nature of the counterexamples and explore alternative metrics or methods to test the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13475
2	preregistration	ollama_remote	glm4:latest	0	0	9874
3	novelty	ollama_remote	glm4:latest	0	0	8419
4	novelty	ollama_remote	glm4:latest	0	0	11344
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12529
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7916
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7577
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7410
9	judge	ollama_remote	glm4:latest	0	0	11707

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 90250 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/cbd4f788bc0d.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/cbd4f788bc0d.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/cbd4f788bc0d.tar.gz` (if generated)